DCT Based JPEG Compression Study

OBJECTIVE

The aim of this experiment is to study how Discrete Cosine Transform (DCT) is used as the mathematical core of JPEG image compression and to qualify the trade off between five size and image quality, multiple compression levels.

- Implement DCT based compression on a grayscale image using the standard JPEG quantisation matrix.
- Apply variable quality factors(Q = 5, 20, 50, 75, 95) to produce the different compression levels.
- Measure and compare image quality using MSE, PSNR and SSIM matrix.
- Compare file sizes between the original and compressed versions at each quality level.
- Visualise DC coefficient energy distribution to understand why compression is effective.

THEORY

What is Image Compression?

Image compression is the process of reducing the size of an image file while retaining as much visual quality as possible. Compression is essential for storage efficiency and fast transmission of images over networks. There are two broad categories:

- Lossless Compression : The original image can be perfectly reconstructed (e.g., PNG). No data is discarded.
- Lossy Compression : Some data is permanently discarded to achieve higher compression ratios (e.g., JPEG). The reconstructed image is an approximation of the original.

JPEG is the most widely used lossy image compression standard in the world, and it is based on the Discrete Cosine Transform.

Discrete Cosine Transform (DCT)

The Discrete Cosine Transform is a mathematical transformation that converts a signal from the spatial domain (pixel values) into the frequency domain (frequency coefficients). It is closely related to the Fourier Transform but uses only real-valued cosine functions.

For an 8x8 block of pixel values, the 2D DCT produces 64 frequency coefficients.

- DC Coefficient : The top most left block. It represents the average brightness of the entire 8x8 block. This is the most important coefficient.
- AC Coefficient : The remaining 63 blocks. They represent increasingly fine detail and texture. Coefficient towards the bottom right of the block represent the highest frequency detail.

The YCbCr Colour Space

JPEG does not compress images in RGB. The image is first converted to YCbCr Colour space. These stand for:

- Y (Luminance) : This represents the brightness of the image. The human eye is highly sensitive to brightness detail.
- Cb (Blue difference Chrominance) : This represents how *blue* the colour is relative to a reference.
- Cr (Red difference Chrominance) : This represents how *red* the colour is relative to a reference.

Since the human eye is far less sensitive to colour detail than to brightness, JPEG applies more aggressive quantisation to the Cb and Cr channels. This discards colour details that the eye cannot easily perceive, achieving higher compression with minimal visible quality loss.

Quantisation

Quantisation is the step where data is permanently lost. Each DCT coefficient is divided by a corresponding value from an 8×8 quantisation matrix and rounded to the nearest integer:

Quantised value = Round(DCT coefficient / Quantisation matrix value)

The quantisation matrix assigns larger divisors to high-frequency coefficients, causing them to round to zero — effectively discarding fine detail. During reconstruction, the coefficients are multiplied back by the matrix (dequantisation), but the rounding error is permanent.

Quality Metrics

1. **PSNR (Peak Signal to Noise Ratio)** : PSNR compares pixel level differences between the original and reconstructed image. It is measured in **decibels (dB)**. A higher PSNR value indicates better quality.

   The formula to calculate PSNR is as follows:

   PSNR = 10log(MAX^2/MSE)

   where,

   MAX is the maximum possible pixel value of the image

   MSE is the *Mean Squared Error*

PSNR Range	Interpretation
> 40 dB	Visually Indistinguishable
30 - 40 dB	Good Quality
20 - 30 dB	Acceptable Quality
< 20 dB	Significant Quality Loss

Typical values for the PSNR in lossy image and video compression are between 30 and 50 dB, provided the bit depth is 8 bits, where higher is better. The processing quality of 12-bit images is considered high when the PSNR value is 60 dB or higher. For 16-bit data typical values for the PSNR are between 60 and 80 dB. Acceptable values for wireless transmission quality loss are considered to be about 20 dB to 25 dB. **The maximum PSNR for 8 bit is 48.131,** for 10 bit is 60.198, for 12 bit is 72.245.

2. **SSIM (Structural Similarity Index)** : SSIM measures similarity in luminance, contrast and structure between two images. It measures how humans perceive quality than PSNR. A higher SSIM indicates more similarity to the original image.

SSIM Range	Interpretation
> 0.95	Excellent Match
0.80 - 0.95	Good Match
< 0.80	Noticeable Quality Loss

EXPERIMENTAL SETUP

Component	Details
Platform	VS Code
Language	Python
Libraries used	NumPy, OpenCV, SciPy, scikit-image, Matplotlib
Quality Levels tested	5, 25, 50, 75, 95
Block size	8x8 pixels
Colour space	YCbCr

METHODOLOGY

Step 1: RGB to YCbCr Conversion

The input RGB image is converted to YCbCr using the standard ITU- R BT.601 transformation matrix. The three channels are compressed separately — Y uses the luminance quantisation matrix, while Cb and Cr use the chrominance matrix.

Step 2: Padding

If the image dimensions are not divisible by 8, the image is padded using edge-replication so that it can be divided evenly into 8×8 blocks. The padding is cropped off during reconstruction.

Step 3: Level Shifting

Each 8×8 block is shifted by −128, centering pixel values around zero (from the range *[0, 255] to [−128, 127]*). This improves the numerical properties of the DCT.

Step 4: 2D DCT

The scipy.dctn(..., norm='ortho') function is applied to each 8×8 block. The norm='ortho' flag ensures orthonormal scaling, preserving energy across the transform.

Step 5: Quantisation

Each DCT coefficient is divided by the corresponding entry in the scaled quantisation matrix and rounded to the nearest integer. The quantisation matrix is scaled based on the quality factor Q using the standard JPEG scaling formula:

- If Q < 50 : Scale = 5000 / Q
- Else : Scale = 200 - 2*Q

Step 6: Restoration

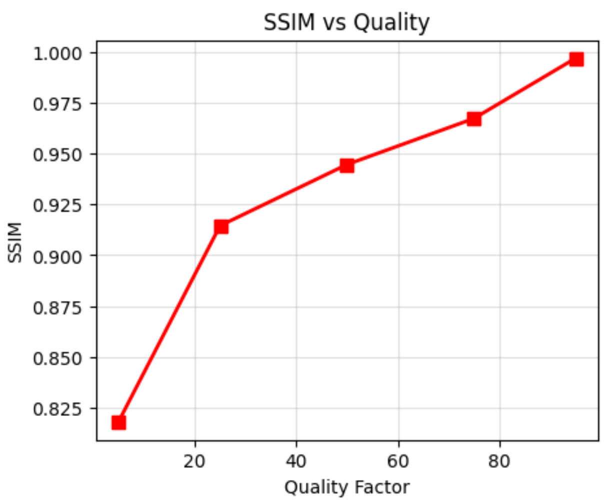
The quantised coefficients are multiplied back by the quantisation matrix (dequantisation), then the Inverse DCT is applied, the level shift is reversed (+128), and values are clipped to [0, 255].

RESULTS

Quality metrics at different quality levels:

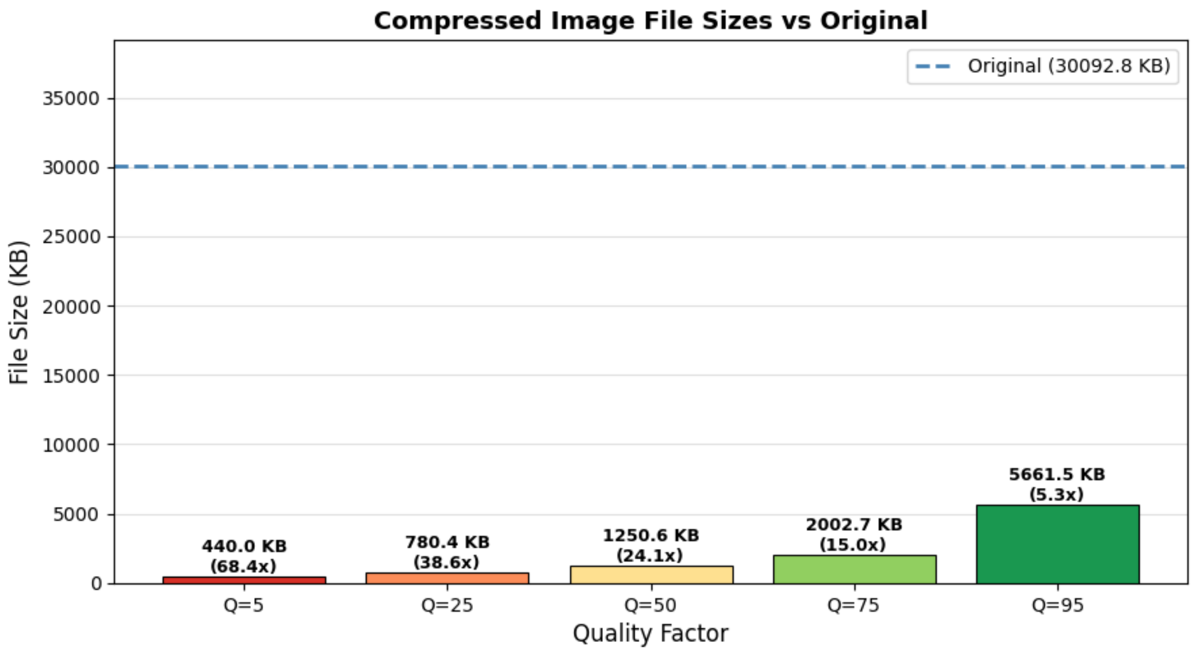
Quality Factor	PSNR %	SSIM	Interpretation
Q = 5	62.1	0.818	Heavy Compression, Visible blocking artefacts
Q = 25	77.9	0.915	Moderate Compression
Q = 50	83.0	0.945	Very Good Quality, Standard JPEG default
Q = 75	88.1	0.967	Excellent Quality, Nearly indistinguishable
Q = 95	100.0	0.997	Near lossless

Note: PSNR here is calculated relative to the maximum PSNR value for 8 bit images, ie, 48.1 dB. It is expressed in percentage relative to that quantity.



File Size Comparison:

Quality Factor	Original Size	Compressed Size	Compressed Ratio
Q = 5	30092.8 KB	440.0 KB	68.4x
Q = 25	30092.8 KB	780.4 KB	38.6x
Q = 50	30092.8 KB	1250.6 KB	24.1x
Q = 75	30092.8 KB	2002.7 KB	15.0x
Q = 95	30092.8 KB	5661.5 KB	5.3x



RESULT ANALYSIS

Effect of Quality Factor on Image Quality :
As the quality factor decreases, quantisation matrix values increase, causing more DCT coefficients — especially high-frequency ones — to round to zero. This produces the characteristic JPEG *"blocking artefact"* where 8×8 block boundaries become visible at low quality settings (Q = 5–15), while at Q = 50 and above the image is visually very close to the original.

YCbCr and Perceptual Compression :
The Y channel contains all the fine detail and sharpness, while Cb and Cr channels are naturally smooth — colour changes slowly across natural images. By applying more aggressive quantisation to Cb and Cr, JPEG discards colour detail that the human eye cannot easily perceive, achieving high compression ratios while preserving perceived sharpness.

DCT Coefficient Visualisation :
After transformation, most image energy concentrates in the top-left (low frequency) coefficients of each block, with bottom-right (high-frequency) values already near zero in smooth regions. Quantisation simply formalises this natural energy compaction by zeroing out the remaining small high-frequency values.

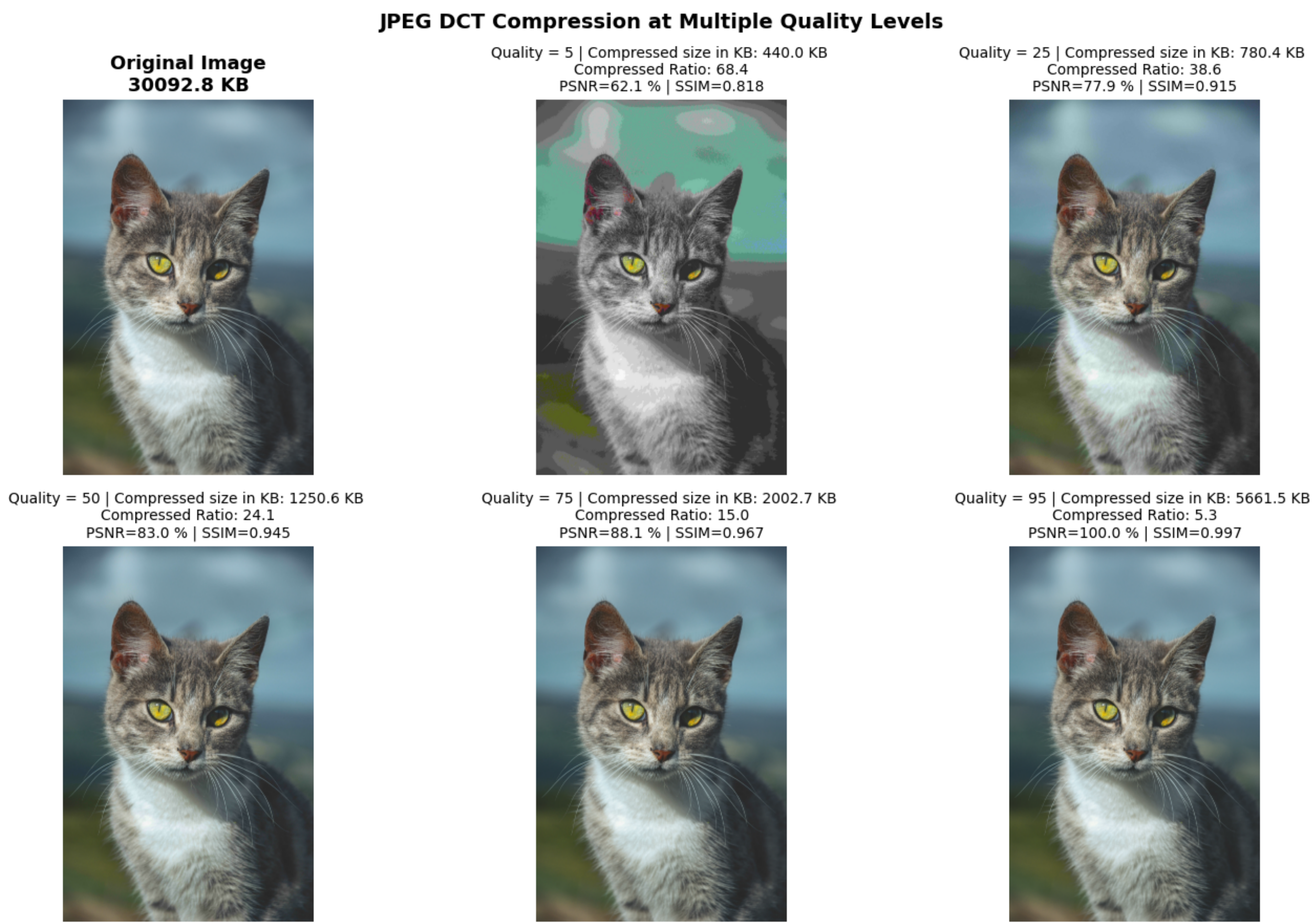
Difference Map and Trade-off :
The greatest pixel-level loss occurs at edges and texture-rich regions, while smooth areas show almost no loss even at low quality settings. This highlights the fundamental trade-off — lower Q gives smaller files but visible degradation, while higher Q gives better quality at the cost of file size.

Our study shows that, Q = 70–85 is the sweet spot.

CONCLUSION

This experiment successfully implemented a complete JPEG-like image compression system from scratch using the Discrete Cosine Transform. The key takeaways are:

- **Quantisation is the core lossy step.** Dividing the DCT Coefficients by the quantisation matrix and rounding permanently discards the high frequency details, which is the fundamental mechanism behind JPEG compression.
- YCbCr conversion allows aggressive compression of colour channels (Cb and Cr) while preserving the brightness (Y), exploiting the human eye's lower sensitivity to colour details.
- The Quality Factor Q directly controls the trade-off between file size and image quality. At Q = 50, the system achieves compression ratios of 24x with a strong balance between size and visual quality.


JPEG DCT Compression at Multiple Quality Levels
Original Image
30092.8 KB
Quality = 5 | Compressed size in KB: 440.0 KB
Compressed Ratio: 68.4
PSNR=62.1 % | SSIM=0.818
Quality = 25 | Compressed size in KB: 780.4 KB
Compressed Ratio: 38.6
PSNR=77.9 % | SSIM=0.915
Quality = 50 | Compressed size in KB: 1250.6 KB
Compressed Ratio: 24.1
PSNR=83.0 % | SSIM=0.945
Quality = 75 | Compressed size in KB: 2002.7 KB
Compressed Ratio: 15.0
PSNR=88.1 % | SSIM=0.967
Quality = 95 | Compressed size in KB: 5661.5 KB
Compressed Ratio: 5.3
PSNR=100.0 % | SSIM=0.997